



Complex Systems

2021/2022

Exam E 22 of July 2022 Time: 2h00

Name:

Number:

This is a **closed books** exam. Any attempt at cheating implies immediate disapproval in the course and the activation of the procedures defined in the disciplinary regulation document of the UC.

Question #	Points	Your score
1	18	
2	12	
3	25	
4	25	
5	20	
	100	

Grader:

Question 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. You may use a concrete example to **help you** to explain the concept.

a) Fixed Point

Your answer:

b) Game of Life.

Your answer:

c) Clustering Coefficient (in a graph).

Your answer:

Question 2 - Comment 12 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) A chaotic systems is different from other types os dynamical systems due to the random nature of its behavior.

Your answer:

Your answer:

- b) A fractal object has a self-similarity dimension greater than its topological dimension.

Your answer:

Your answer:

Question 3 - Simulate 25 points

Consider the system described by the following difference equation:

$$x_{n+1} = x_n^3$$

Identify the fixed points of the system, if any, describe their stability and state the long term behavior of the system. Draw the corresponding phase line.

Your answer:

Your answer:

Question 4 - Explain 25 points

In class we presented and studied a model for a **predator-prey** ecosystem based on differential equations. Present the model and explain its components. In what way an agent-based approach or a network-based approach is more appropriate? Justify. Do not forget to describe these two models.

Your answer:

Your answer:

Question 5 - [Model](#) 20 points

Imagine a population in which the organisms may be at three different stages of development: larvae, young adults and adults. Suppose that 1% of larvae grow into young adults every year, and 30% of young adults survive to become (older) adults. Furthermore, young adults have an average of 104 offspring each year, while older adults have an average of 160 offspring each year. Based on these relations model the dynamics using **difference equations**. Supposing that initially you have only 100 larvae, compute the value of larvae, young adults and adults **after two years**.

Your answer:

Your answer: